

HEALTH CARE ANALYSIS DASHBOARD REPORT

Introduction:

The healthcare sector relies heavily on research funding to advance medical innovation and improve public health outcomes. This project uses Tableau Public to visualize and analyse data from the RCDC (Research, Condition, and Disease Categorization) Dataset.

The analysis focuses on understanding how research funding is distributed across various disease areas, how it has changed over the years, and how it relates to mortality and prevalence rates. By applying different visualization techniques and trend models, the project aims to uncover meaningful insights that highlight key funding trends within the U.S. healthcare research landscape.

Methodology:

➤ Dataset Overview:

- The dataset used for this project is the RCDC (Research, Condition, and Disease Categorization) Healthcare Funding Dataset, which provides funding information across multiple research and disease areas from 2008 to 2024.
- It includes details such as funding amount, funding type, year of funding, Research/Disease Categories, Mortality rate and Prevalence of USA.

➤ **Data Cleaning and Preparation:**

- Removed irrelevant or empty rows and corrected null/missing values.
- Replaced special symbols such as “*” and “-” with zero (0).
- Checked for negative or invalid values in funding and mortality columns
- Standardized column names for easier recognition in Tableau.
- Created a new column for “State” (randomly assigned U.S. states) to support geographic visualization.

➤ **Data Transformation in Tableau:**

- Pivoted year columns from 2008 – 2024 into one column names Year
- Created a new field “Funding Type” to classify Regular Funding, ARRA Funding and Estimated Funding.
- Joined the Category Info sheet containing Research Categories using Research/Disease Area as the key.
- Added calculated fields such as Running Total, Yearly Growth %, and Funding Difference.

➤ **Data Visualization and Modelling:**

- The cleaned and structured dataset was used to create several visualisation.
- Bar, Line and Dual Line Charts for Time trends.
- Tree Map, Heat Map and Funnel Chart for distribution analysis.
- Whisker Chart (Box Plot) for variability
- Scatter Plot for correlation between Funding vs Mortality.
- Symbol Map for geographic funding visualization.
- Trend Analysis using Linear, Logarithmic, Exponential, and Polynomial models.

Key Findings:

➤ Overall Funding Growth:

- Total healthcare research funding has shown a steady upward trend from 2008 to 2024.
- Clinical Research has a steady funding over the years.
- After 2015, funding growth accelerated in some of the research areas.
- The funding for pregnancy spiked in year 2018 and it didn't have any funding until then.

➤ Funding Trends across research areas:

- Heart Disease and Autism Research showed strong funding stability, with noticeable increases in the past five years.
- **Sleep Research** and **Down Syndrome** funding remained comparatively lower, reflecting limited allocation to specialized domains.
- **Cancer Research** consistently received the **highest funding**, rising from around 5k in 2008 to over 8k by 2024.

➤ Funding Distribution by Category (2024):

- In 2024, Clinical Research, Genetics, and Neurosciences dominated funding allocations.
- Areas like Prevention and Behavioural Sciences received moderate funding, whereas smaller fields such as Brain Disorders and Biotechnology had lower shares.

➤ Disease specific insights:

- Cancer experienced both steady funding growth and strong predictive trend fits

- Pregnancy and Infertility Research displayed fluctuating funding patterns, highlighting variable investment over time.
 - HIV/AIDS and Vaccine-related Research showed periodic spikes, aligning with major global health initiatives.
- **Geographic Funding Patterns:**
- States such as Maryland, Washington and New Jersey received the highest research funding about 400k dollars.
 - States like North Carolina and Virginia received significantly lower allocations, indicating regional funding disparities.
- **Trend Model Findings:**
- The linear model shows a steady growth at a constant rate for cancer research area
 - The Exponential Model shows a rapid rise in recent years, indicating accelerating investment.
 - The polynomial model captures temporary dips followed by recovery.
 - The Logarithmic Model shows less fit due to continuous upward growth in cancer funding.
- **Funding Distribution and Variability:**
- **Whisker Charts** indicate wide variation in funding among different research areas.
 - **Cancer, HIV/AIDS, and Rare Diseases** show higher variability — signaling targeted, high-impact funding initiatives.

- Areas like **Nutrition and Mental Health** maintain narrow funding ranges, reflecting stable but modest allocations

➤ **Correlation Insights:**

- The Scatter Plot between funding and 2019 U.S. mortality rates suggests a weak negative correlation, meaning higher funding is generally associated with lower mortality in major research domains.

Actionable Insights:

➤ **Strategic Funding Allocation:**

- Increase funding for underrepresented research areas such as Sleep Disorders, Down Syndrome, and Infertility Research, which show lower and inconsistent funding across the years.
- Maintain or further expand funding for Cancer and Heart Disease Research, which demonstrate strong growth and tangible health outcomes.

➤ **Geographic Investment Optimization:**

- Focus future funding efforts in states with lower allocations like Virginia and North Carolina to reduce regional research disparities.
- States such as Washington, Maryland and New Jersey already receive high investments; expanding partnerships with underfunded regions can encourage nationwide research balance.

➤ **Trend-Based Resource Planning:**

- Linear and exponential trend models reveal continued growth in cancer's research investment through 2024 — stakeholders can plan long-term funding commitments based on these projections.
- Polynomial models highlight short-term fluctuations

➤ **Strengthen Correlation-Driven Policies:**

- The scatter plot analysis indicates a weak negative correlation between funding and mortality rates, suggesting that sustained investment can gradually reduce mortality
- Policymakers should direct higher funding toward diseases with both high mortality and low current funding, to maximize health impact.

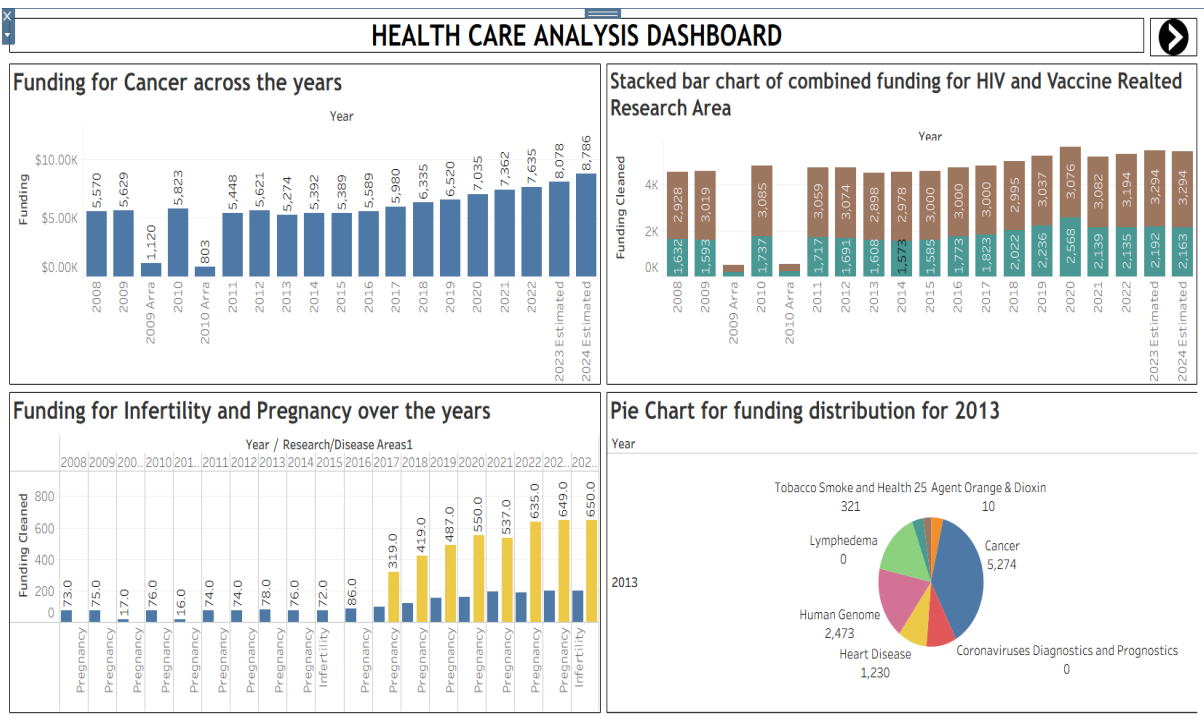
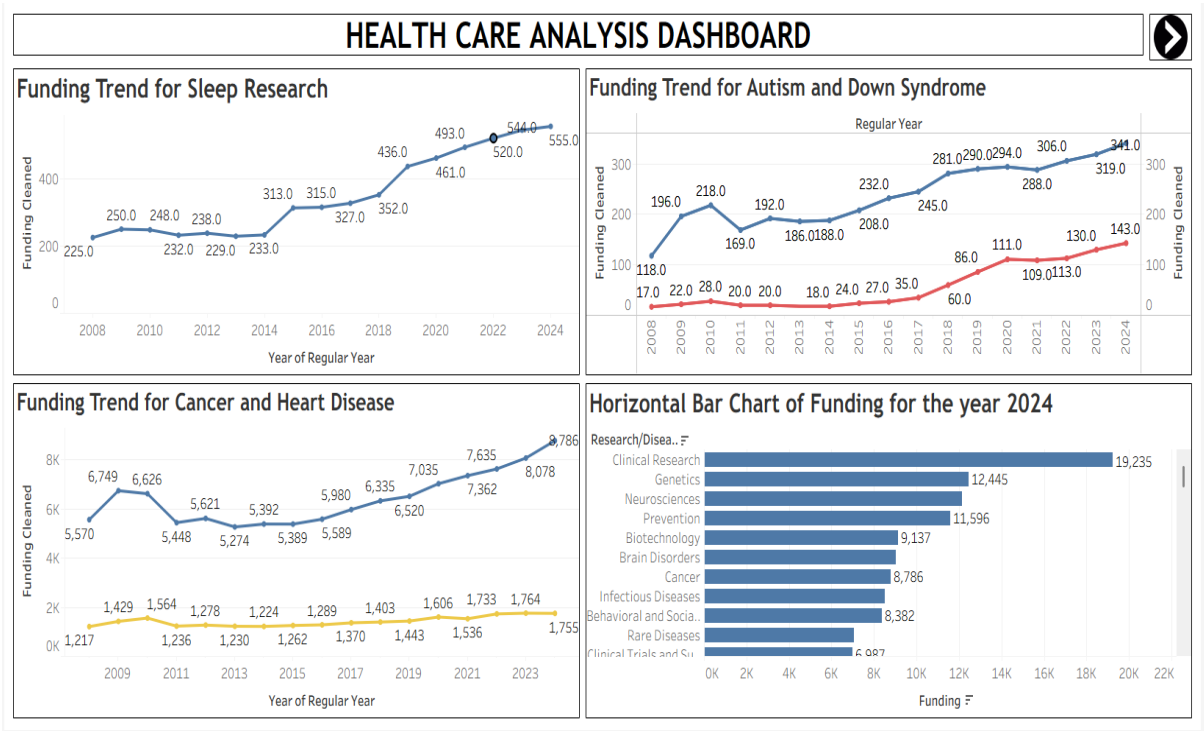
➤ **Funding Stage Evaluation:**

- Regular funding remains the backbone of research support; however, ARRA in 2008 and 2009 and Estimated funding in 2023 and 2024 show significant growth potential
- Continued evaluation of these funding stages will help determine which financial provides the best outcomes per dollar spent.

➤ **Improving Efficiency Through Data-Driven Insights:**

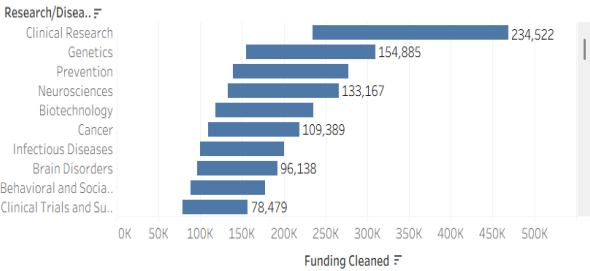
- Use running sum and window sum analyses to track cumulative funding efficiency and identify stagnating projects.

Visualisations:

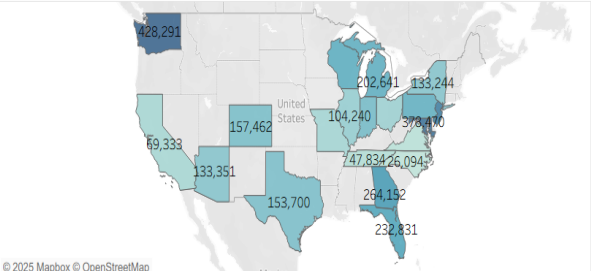


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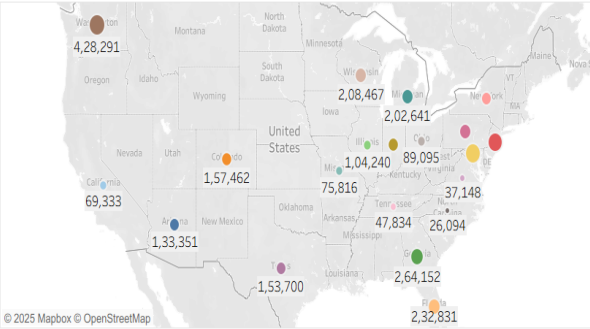
Gantt Chart for Research funding



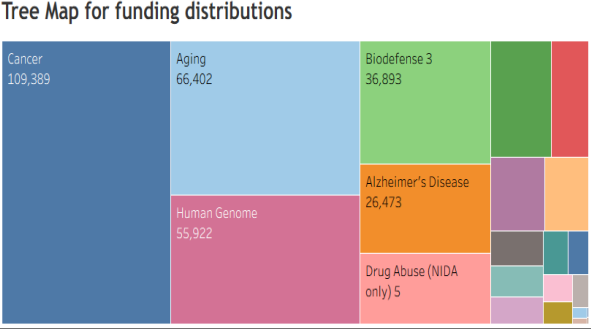
Filled Map showing funding distributions



Symbol Map for funding from each state

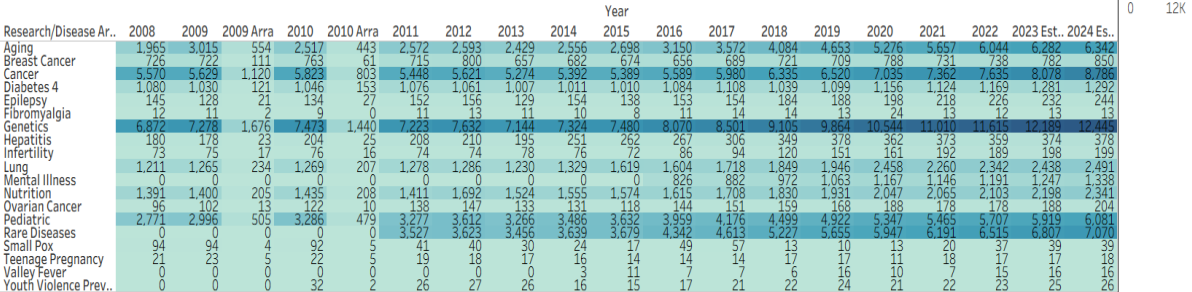


Tree Map for funding distributions

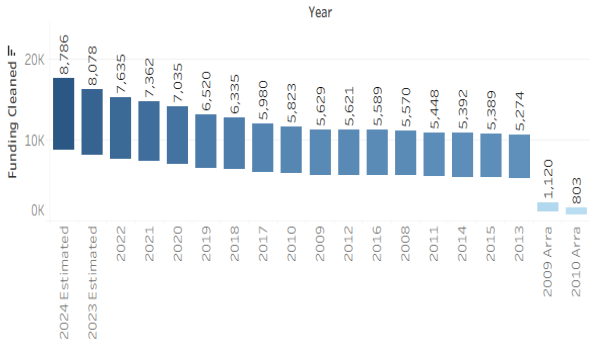


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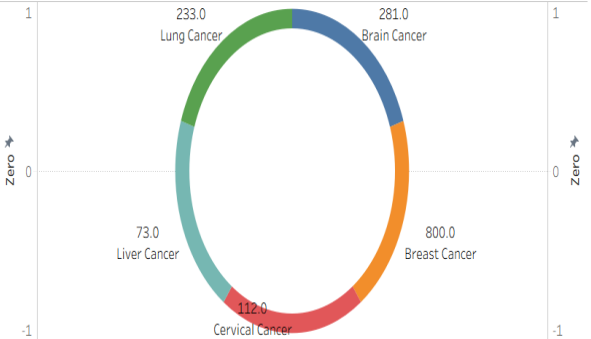
Heat Map showing Funding Intensity



Water Fall Chart showing funding for Cancer

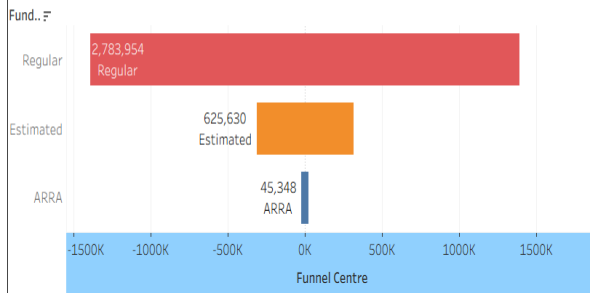


Donut Chart representing funding of various cancer disease

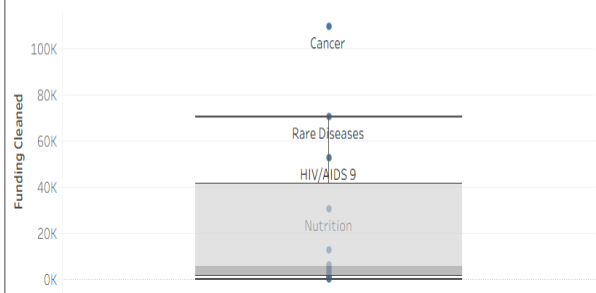


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Funnel Chart showing funding stages



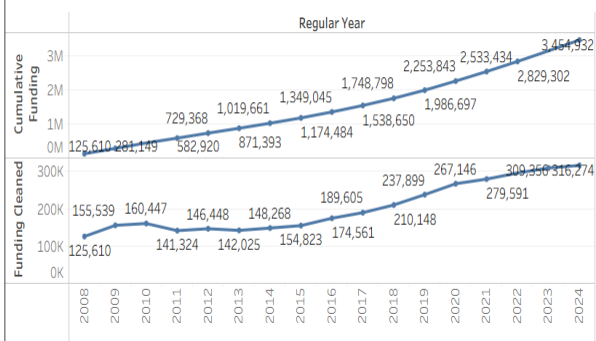
Whisker Chart



Scatter Plot

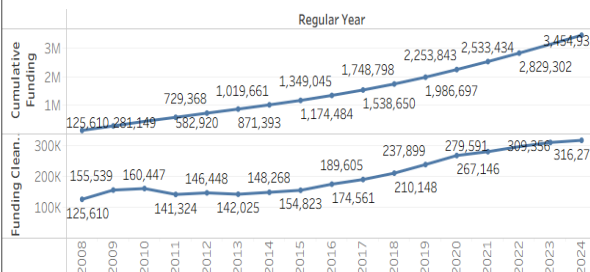


Running Sum

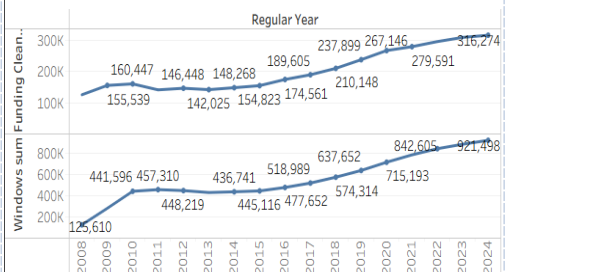


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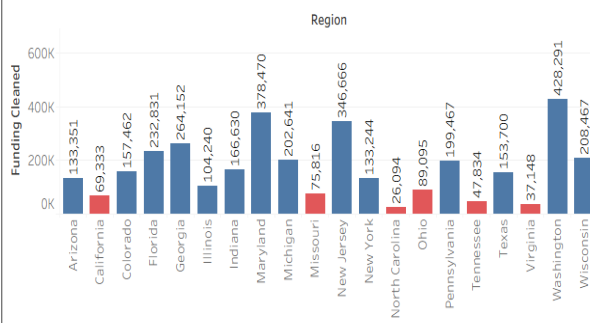
Running Sum



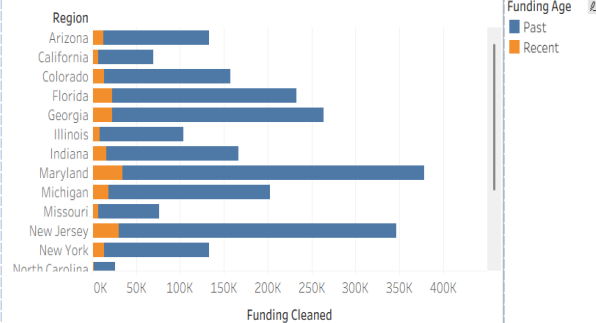
Windows Sum



Comparison



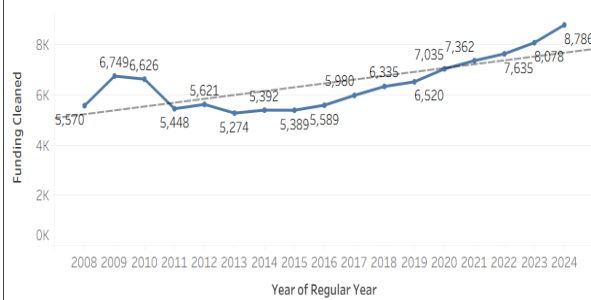
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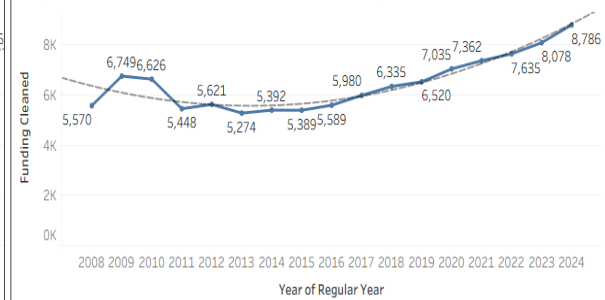


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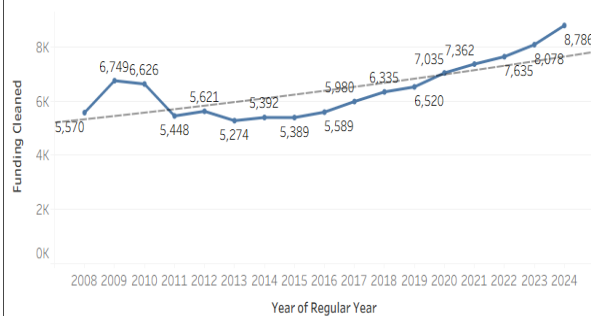
Logarithmic Trend Line for Cancers Funding



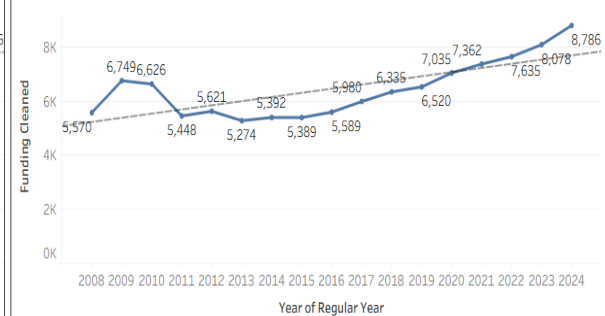
Polynomial Trend Line for Cancer's funding



Exponential Trend Line for Cancers's funding



Linear Trend Line for Cancer's funding



Conclusion:

- The project analysed U.S. healthcare research funding trends using the RCDC dataset in Tableau.
- Funding showed a steady upward trend from 2008 to 2024, with major spikes during the Estimated years (2023–2024).
- Genetics, Clinical Research and Prevention received the highest funding, while smaller areas like Poly Cystic Ovary Syndrome, Human foetal Tissue, Neck Pain, Vaginal Cancer etc remained underfunded.
- Trend models highlighted overall positive growth and cyclical variations in funding.
- The study ensures the value of data-driven decision-making for optimizing healthcare research investments and promoting good funding distribution.